



PRT661 – Data Science Practice

Assessment 1 – Project Proposal and Design

DataGuard: Trust-Aware Anomaly Detection for Untrusted Data Pipelines

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1. Project overview and objectives

Monitoring systems are designed to raise alarms when unusual behaviour caught in the whole process. But data can look unusual for different reasons like something really happened or something broke in the pipeline. In such cases the system cannot tell the reason for the alarm give might give us false alarm.

For example, if a city has five air quality sensors and three of them stop working, the average pollution reading will change even though the air did not. A normal monitoring system reports this as an improvement which is false information. Such cases leads to data failure.

DataGuard addresses this. The system runs two detection layers within a single data pipeline. Layer 1 checks whether the data itself is healthy, whereas Layer 2 looks for genuinely unusual events in the real-world. A fusion component gives every Layer 2 alert a trust score based on Layer 1. Alerts raised while the data was broken are held back for the review instead of being escalated.

Objectives:

1. Build an automated pipeline that collects and stores global air quality measurements.
2. Detect data quality problems.
3. Detect genuine air pollution events using unsupervised methods.
4. Measure how much the trust score reduces false alerts.
5. Develop a publicly accessible monitoring dashboard.

2. Problem statement

OpenAQ collects air quality measurements from thousands of sensors worldwide and publishes them in one place. OpenAQ does not own or operate these sensors. Governments, research groups and other organizations own them. OpenAQ just collects whatever those sources publish and publishes it into a single uniform format.

The data changes and degradation are documented and verifiable. OpenAQ retired its version 1 and version 2 endpoints on 31 January 2025, and those endpoints now return an HTTP 410 error [1]. OpenAQ also publishes a delivery commitment stating that files are written 72 hours after the end of day in the location's timezone [2], which means late files can be measured against a stated promise rather than a threshold we invented.

Sensor-level failures are directly observable in the data. A sensor may report the same value for weeks, report a negative concentration, or stop reporting entirely. Air pollution cannot be negative, and real air never holds at exactly one number for weeks. These readings are self-evidently broken. And, genuine pollution events such as bushfire smoke, dust storms and industrial incidents appear as large changes in the same measurements.

3. Target users

Primary: environmental monitoring and data platform teams. These teams publish air quality information to the public. They need to know when their data has degraded before people act on it. At present, they usually find out only when someone complains.

Secondary: downstream users of air quality data. Researchers, public health agencies, city planners and app developers all use this data. They don't know why three sensors in a city stopped reporting last Tuesday, or that a station was moved. Their conclusions can be wrong without them realising it.

4. Business case

Poor data quality is expensive. Gartner estimates it costs organisations an average of \$12.9 million per year, and identifies inconsistency between sources as one of the biggest data quality problems organisations report [4].

There are two main costs. First, teams waste time investigating alerts caused by pipeline faults rather than real events. Second, too many false alerts cause alert fatigue. Once people stop trusting alerts, they ignore the real ones too.

DataGuard checks the health of the data behind every alert. The measure of success is high detection with fewer false alerts.

5. Why this is a Data Science project

The project addresses two Theme 3 applications, which are operational anomaly detection and intelligent monitoring and alerting systems. It is a data science project rather than a machine learning application for four reasons.

First, the training data does not already exist. The source contains pollution measurements, not a list of which days were healthy. We build that dataset ourselves. For every station and day, we calculate quality measures such as how many readings arrived, how many were missing, whether any value repeated unchanged, and how late the file was. Those measures become the features the detector learns from.

Second, there is no ground truth. No one provides a list of which days are broken. We must define what an anomaly means, then validate our approach using documented platform changes, self-evident sensor failures, and controlled synthetic problems.

Third, most of the work is data engineering and analysis. Collecting the data, designing the storage, reconciling differences between thousands of providers, and building useful features.

Fourth, the system supports human decisions rather than making predictions. It helps a person decide whether an alert deserves attention, based on the quality of the evidence behind it.

6. Initial architecture

DataGuard — Trust-Aware Anomaly Detection Architecture

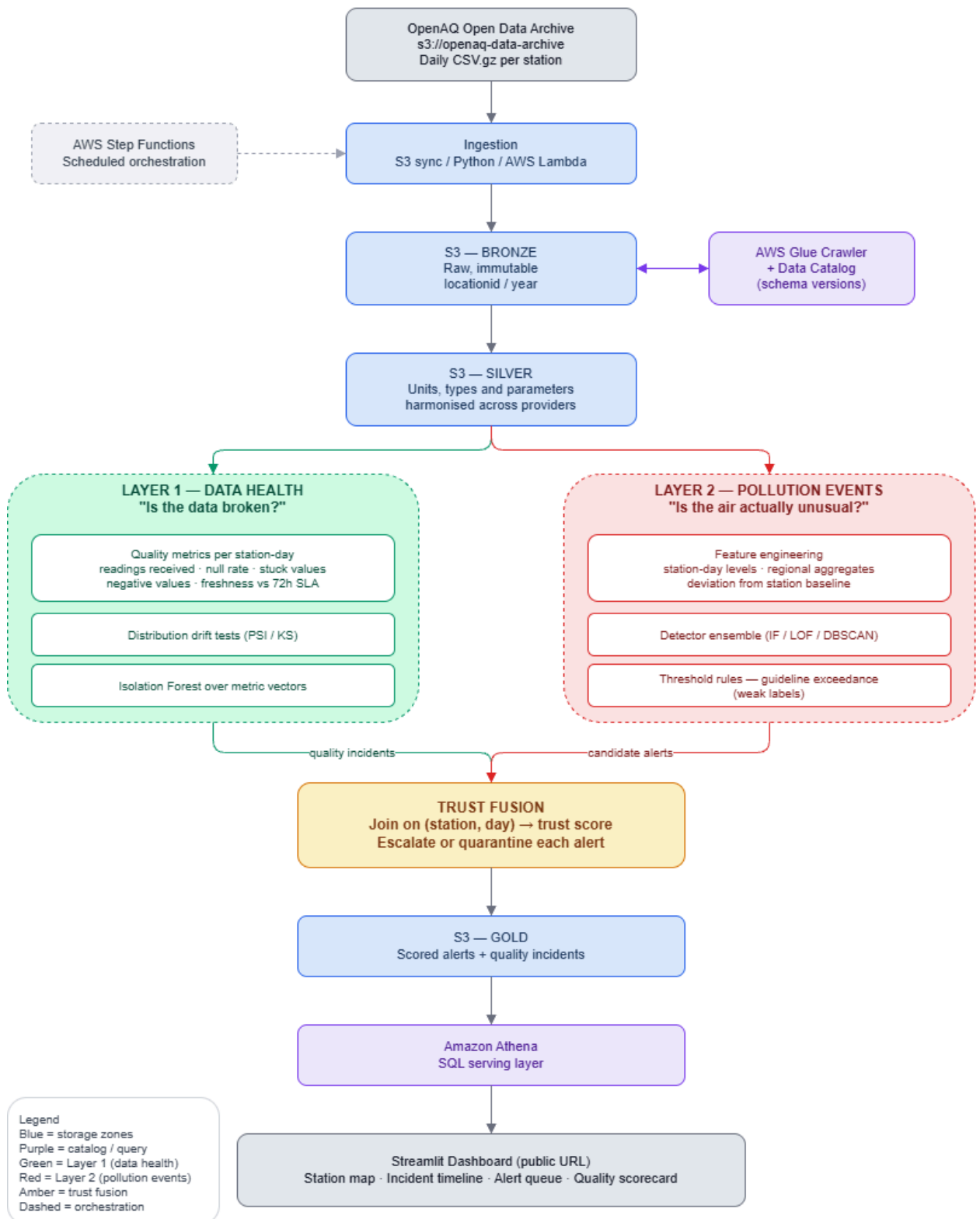


Figure1. DataGuard system architecture

Raw measurement files are stored unchanged in an S3 bronze zone, organised by location and year. An AWS Glue Crawler records the structure of each file, which lets us detect changes in the data over time.

A silver zone holds cleaned data with consistent units and types across all providers. Both detection layers read from the silver zone and write their results to a gold zone.

The gold data is queried through Amazon Athena and displayed on a Streamlit dashboard. AWS Step Functions runs the pipeline automatically on a schedule.

7. Workflow plan

The system will be based on batch data processing, not streaming data. Source data is published monthly with an approximate two-month lag [1].

Phase	Deliverable	Gate
Foundation	Ingestion automated; bronze zone populated; Glue Catalog active	Data queryable via Athena
Conformance	Silver zone; units and types consistent across providers	Single queryable table
Layer 1	Quality metrics per station-day; drift tests; anomaly model	Detects known failures unprompted
Layer 2	Pollution event features; detector ensemble	Ranked anomaly output
Fusion	Trust scoring; dashboard deployed	Public URL live
Consolidation	Documentation, final report, presentation	Submission

Table 1. Delivery phases

8. Task allocation and team structure

Member	Role	Responsibility
Aadarsh Ghimire	Data engineer	Ingestion pipeline, raw/silver/gold zone design, partitioning strategy, Glue Crawler & Data Catalog, job orchestration

Orchid Shrestha	Data quality engineer	Layer 1 data quality metrics, schema drift testing, quality anomaly model
Sandesh Prasad Paudel	Machine learning engineer	Layer 2 feature engineering, anomaly detector ensemble, model evaluation
Shuvechcha Pun	Analytics and visualisation	Gold tables query design, Athena queries, dashboard build, station mapping
Sandesh Shahi	Project lead	Fusion layer, repository governance, diagrams, integration, reporting

Table 2. Role allocation.

9. Initial risk analysis

#	Risk	L	I	Mitigation
R1	Reconciling thousands of providers takes longer than planned	M	H	Five-day timebox; narrow to a single region if exceeded
R2	AWS Academy credits exhausted	M	H	Mandatory partition filtering; weekly spend review; local DuckDB fallback for serving
R3	Fragmented individual AWS accounts impede integration	M	H	Infrastructure as reproducible scripts; one nominated integration account
R4	No ground truth for Layer 2 evaluation	H	M	Rule-based weak labels; Precision@K; independent review by two members
R5	Concept not understood by stakeholders	M	H	Worked example precedes architecture in all communication
R6	Integration deferred to final weeks	H	H	Weekly integration checkpoint; five-day branch limit
R7	Scope expansion beyond team capacity	H	H	Mid-project go/no-go gate; final week reserved as buffer

Table 3. Initial risk register. L = likelihood, I = impact.

10. Ethics, privacy and security considerations

Privacy. The data contains environmental measurements from fixed monitoring stations. It contains no personal information, and no individuals can be identified from it. This substantially reduces privacy risk compared with human-generated datasets.

Public safety. Air quality data informs real health decisions. Suppressing a genuine pollution alert could cause harm. For this reason, quarantined alerts are always retained and displayed for human review, never deleted or hidden. The system reduces noise; it does not withhold information.

Fairness. Layer 2 may flag anomalies associated with particular sensor operators, agencies or countries. Such output reflects data collection processes, not proven negligence. All results are presented as leads for investigation, never as judgements about any organisation.

Appendix

Github link: <https://github.com/shahsandesh30/DataGuard>

Jira Link:

<https://aadarshghimire524.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog?epics=visible&atlOrigin=eyJpIjoiZmNjMDQwMjViOTE2NGY1MDg5ZmUwNjhlNzgxY2MwYmEiLCJwIjoiYiJ9>

(reem.sherif@cdu.edu.au has been added as a viewer in the Jira Space)

References

- [1] OpenAQ, "About the API." <https://docs.openaq.org/about/about>
- [2] OpenAQ, "About Open Data on AWS." <https://docs.openaq.org/aws/about>
- [3] Registry of Open Data on AWS, "OpenAQ." <https://registry.opendata.aws/openaq/>
- [4] Gartner, "Data Quality: Why It Matters and How to Achieve It." <https://www.gartner.com/en/data-analytics/topics/data-quality>
- [5] F. T. Liu, K. M. Ting and Z. Zhou, "Isolation Forest," Eighth IEEE International Conference on Data Mining, 2008, pp. 413-422. Applied in Layer 1 to detect abnormal patterns in quality metrics.